Bang-time perturbations and redshift drift in LTB cosmology: regularity, distance compensation, and light-cone response

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Abstract

We calculate the redshift-drift response to a radially varying bang time in Lemaître–Tolman–Bondi (LTB) models by propagating neighboring light cones. An exact parabolic dust solution yields an explicit variational equation and reveals that the initially studied hump-shaped bang-time profiles have shell crossings near the bang for either sign of their amplitude. Replacing them with monotone profiles gives a globally shell-crossing-free expanding parabolic family for positive amplitude. Its equal-distance drift response is positive at all 15 sampled redshifts $0.1 \le z \le 1.5$, opposite to the diagnostic hump profiles. An exact flat Λ-LTB family with $(\Omega_m, \Omega_\Lambda) = (0.3, 0.7)$ gives the same sampled sign. We also derive a linear curvature compensation that enforces $\delta D_L(z) = 0$ locally; its center-regular solution grows sharply toward the background angular-distance maximum, where the inverse equation is singular. These are model-family results, not a measurement or a universal sign theorem. The numerical integrations, independent derivative checks, and figures are reproducible.

1 Scope and relation to earlier work

Redshift drift in spherically symmetric spacetimes and its use with distance data were established by Uzan, Clarkson and Ellis [1]. Yoo, Kai and Nakao developed the LTB drift calculation and studied void models [2]. Exact growing and decaying density modes in LTB dust have been analyzed by Sussman [3]. The inverse problem for LTB observables has additional existence and regularity conditions [4]. We therefore claim neither a new general drift equation nor a unique observational signature. The purpose here is to correct an endpoint-formula error in an earlier unpublished draft, spell out an explicit parameter derivative, and give a small reproducible test case.

In units $c = G = 1$, the LTB metric and evolution equation are

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \frac{R'(t,r)^2}{1+2E(r)}\mathrm{d}r^2 + R(t,r)^2\mathrm{d}\Omega^2, \tag{1}$$

$$\dot{R}^2 = \frac{2M(r)}{R} + 2E(r), \qquad R(t_{\mathrm{B}}(r), r) = 0. \tag{2}$$

Dots and primes denote partial derivatives at fixed r and t, respectively. A radial coordinate choice removes one of M, E, t_{B} as an independent function. A nonconstant t_{B} represents the decaying mode in the conventional LTB dust mode decomposition [3]; its additive constant sets the time origin. Throughout this paper the mass-coordinate gauge and the profiles below are specified explicitly.

The calculation asks five concrete questions: how to propagate the drift of the same source; how comparison at fixed redshift differs from equal distance; whether the response's sign survives changes of bang-time shape and regularity conditions; whether a curvature mode can keep the full distance curve

fixed; and whether these effects persist on a matter-plus-Λ background. The equations and computed results below answer these questions within precisely stated model families. Observational viability requires a separate fit to early- and late-time data.

2 Drift on neighboring radial light cones

Put $S(r) = \sqrt{1+2E(r)}$, $A = R'/S$, $B = \dot{R}'/S$, and $C = \ddot{R}'/S$. On an incoming radial ray from a central, comoving observer at $r = 0$, the emission time $t(r)$ and observed redshift $z(r)$ satisfy

$$\frac{\mathrm{d}t}{\mathrm{d}r} = -A, \qquad \frac{\mathrm{d}z}{\mathrm{d}r} = (1+z)B, \qquad t(0) = t_0, \quad z(0) = 0. \tag{3}$$

Compare two rays received at times separated by δt_0, from the *same comoving emitter* r. Their time separation $T(r) = \partial t(r; t_0)/\partial t_0$ obeys $T' = -BT$, $T(0) = 1$, hence $T = 1/(1+z)$. Let $Z(r) = \partial z(r; t_0)/\partial t_0$ be drift per unit central proper time. Differentiating Eq. (3) with respect to t_0 at fixed r gives the exact propagation equation

$$Z' = BZ + (1+z)CT = BZ + C, \qquad Z(0) = 0, \tag{4}$$

or, equivalently,

$$\boxed{Z(r) = (1+z(r)) \int_0^r \frac{C(\bar{r})}{1+z(\bar{r})}\,\mathrm{d}\bar{r}.} \tag{5}$$

This result carries the information along the ray. In a homogeneous FLRW limit it reduces to $Z = (1+z)H_0 - H(t_e)$. In a general LTB geometry, replacing Eq. (4) by $(1+z)H_0 - \dot{R}'/R'|_e$ is not justified: the gradient of expansion along the ray contributes. For an isotropic central observer, reciprocity gives

$$D_L(r) = (1+z(r))^2 R(t(r), r). \tag{6}$$

3 Linear response and what "fixed distance" means

Let p denote a model parameter and $\boldsymbol{y} = (t, z, Z)$. With $\boldsymbol{y}' = \boldsymbol{F}(\boldsymbol{y}, r, p)$ from Eqs. (3) and (4), its first-order variation $\boldsymbol{u} = \partial_p \boldsymbol{y}|_r$ is specified, without finite differences, by

$$\boldsymbol{u}' = \underbrace{\begin{pmatrix} -A_t & 0 & 0 \\ (1+z)B_t & B & 0 \\ B_t Z + C_t & 0 & B \end{pmatrix}}_{\boldsymbol{F}_{\boldsymbol{y}}} \boldsymbol{u} + \begin{pmatrix} -A_p \\ (1+z)B_p \\ B_p Z + C_p \end{pmatrix}, \qquad \boldsymbol{u}(0) = \boldsymbol{0}. \tag{7}$$

All partial derivatives in this equation are at fixed t, r; in particular, A_p includes any parameter dependence in S. Equation (7) is the analytic variational equation used to define the linear response; the public numerical example evaluates its derivative by symmetric finite differences and checks step convergence.

At a given emitter coordinate the distance response is

$$\partial_p D_L|_r = (1+z)^2(R_t t_p + R_p) + 2(1+z)R z_p, \qquad \frac{\mathrm{d}D_L}{\mathrm{d}r} = 2(1+z)z_r R + (1+z)^2(R_t t_r + R_r). \tag{8}$$

Here R_p and R_r are partial derivatives, whereas $\mathrm{d}D_L/\mathrm{d}r$ is a total derivative along the ray. If $\mathrm{d}D_L/\mathrm{d}r \neq 0$, implicit differentiation yields the response at a *single equal luminosity distance*,

$$\boxed{\left.\frac{\partial Z}{\partial p}\right|_{D_L} = Z_p|_r - \frac{Z_r}{\mathrm{d}D_L/\mathrm{d}r}\,\partial_p D_L|_r.} \tag{9}$$

Table 1: Diagnostics produced by `python analysis.py`. Extremes and convergence refer to the 15 sampled distances, not a continuum bound.

w	$\min \partial_\epsilon Z\|_{D_L}$	$\max \partial_\epsilon Z\|_{D_L}$	max. change on halving h
0.8	-0.328043	-0.001464	1.39×10^{-7}
1.2	-1.282333	-0.001490	1.81×10^{-7}

For comparison, $\partial_p Z|_z = Z_p|_r - (Z_r/z_r)z_p|_r$. Neither condition requires $D_L(z)$ to be identical between the two models. Keeping the *whole function* $D_L(z)$ fixed would require a compensating functional perturbation in M or E and is a distinct inverse problem. Equations (7)–(9) also show that a zero-crossing location cannot be inferred from the background profile alone without specifying the allowed perturbation and compensation.

4 Exactly specified illustrative model

Set $t_0 = 1$, $E(r) = 0$, $M(r) = 2r^3/9$, and $t_{\rm B}(r) = \epsilon f_w(r)$, with

$$f_w(r) = r^2 \exp[-(r/w)^2], \qquad w \in \{0.8, 1.2\}. \tag{10}$$

All coordinates and drift values in this section are dimensionless. The two choices of w are perturbation shapes on the *same homogeneous background*, not two realistic void profiles. The even behavior at $r = 0$ makes the center smooth. The exact expanding solution, with $\tau = t - \epsilon f_w(r)$, is

$$\begin{aligned} R &= r\tau^{2/3}, \\ R' &= \tau^{2/3} - \frac{2r\epsilon f_w'}{3\tau^{1/3}}, \quad \dot{R}' = \frac{2}{3\tau^{1/3}} + \frac{2r\epsilon f_w'}{9\tau^{4/3}}, \\ \ddot{R}' &= -\frac{2}{9\tau^{4/3}} - \frac{8r\epsilon f_w'}{27\tau^{7/3}}. \end{aligned} \tag{11}$$

The expansion branch requires $\tau > 0$; the code checks $R' > 0$ and monotonic z, D_L on every sampled light cone. At $\epsilon = 0$, the model is Einstein–de Sitter (EdS) and has the exact check

$$Z_{\rm EdS}(z) = \frac{2}{3}\big[(1+z) - (1+z)^{3/2}\big]. \tag{12}$$

At $\epsilon \neq 0$ we hold M, E, t_0 fixed, integrate to $z = 1.6$, and compare drift values at the *baseline* $D_L(z)$ for $z = 0.1, 0.2, \ldots, 1.5$. The derivative uses $[Z(+h) - Z(-h)]/(2h)$ at $h = 0.002, 0.001, 0.0005$. The plotted result uses $h = 0.0005$.

At $\epsilon = 0$, the full null-cone solution provides further independent checks:

$$t(r) = (1 - r/3)^3, \qquad 1 + z(r) = (1 - r/3)^{-2}, \qquad D_L(z) = 3\big[(1+z) - \sqrt{1+z}\big]. \tag{13}$$

The source radius corresponding to a baseline redshift is $r(z) = 3[1 - (1+z)^{-1/2}]$. It is this baseline distance and radius, rather than the perturbed model's redshift label, that define the equal-distance sample.

The maximum absolute baseline error against Eq. (12) on this grid is 4.4×10^{-13}. An independent two-observer-time finite difference of redshift at fixed $r = 0.8$, with $\epsilon = 0.001$ and time step 10^{-5}, agrees with the propagated drift to 3.3×10^{-11}. For $\epsilon = 0.001$, the first-order maximum fractional distance change at fixed baseline z is approximately 2.20×10^{-4} ($w = 0.8$) or 3.78×10^{-4} ($w = 1.2$). Thus the equal-distance operation is numerically distinct from holding the complete distance relation fixed.

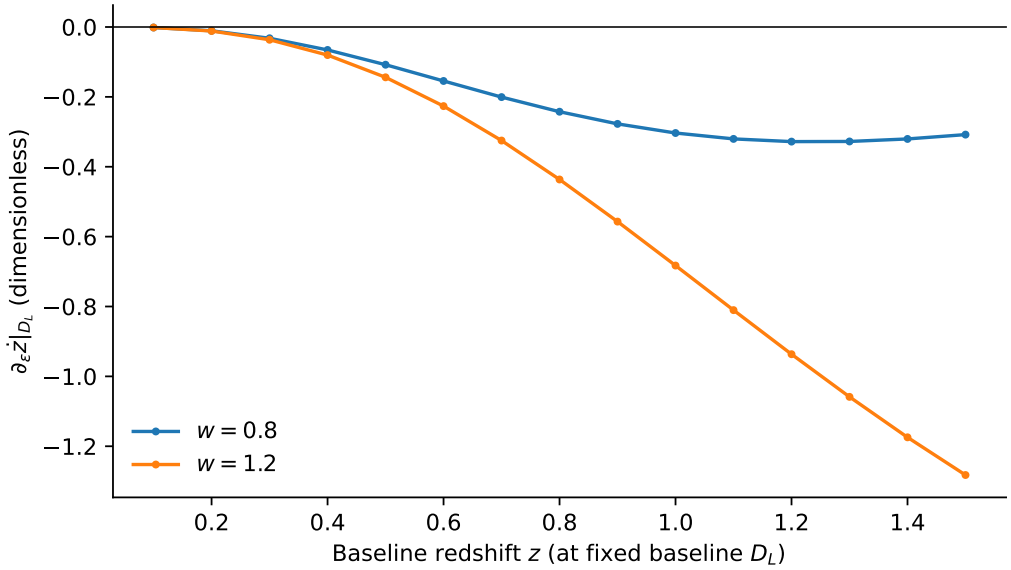


Figure 1: Dimensionless bang-time response at equal D_L for the two specified f_w. The horizontal coordinate labels the *baseline* redshift. Lines guide the eye between 15 sampled points; no observational uncertainties are implied.

Table 2: Response derivatives at baseline $z = 1$ (dimensionless). The last column applies the FLRW endpoint shortcut at fixed z and is not a valid LTB drift calculation.

w	exact, fixed z	exact, equal D_L	endpoint shortcut, fixed z
0.8	-0.158830	-0.303455	$+0.164275$
1.2	-0.426856	-0.682806	-2.670339

Table 1 and Fig. 1 support the narrow statement that these two examples have no sign change at the sampled distances. They do not establish a sign theorem for arbitrary f, nonparabolic backgrounds, or compensated models.

The linear variational calculation in Appendix A agrees with the $h = 0.0005$ equal-distance finite difference to at most 4.7×10^{-8} for $w = 0.8$ and 6.1×10^{-8} for $w = 1.2$ on the grid. Figure 2 and Table 2 make the comparison choice and endpoint-formula error visible rather than hiding them inside a single response curve.

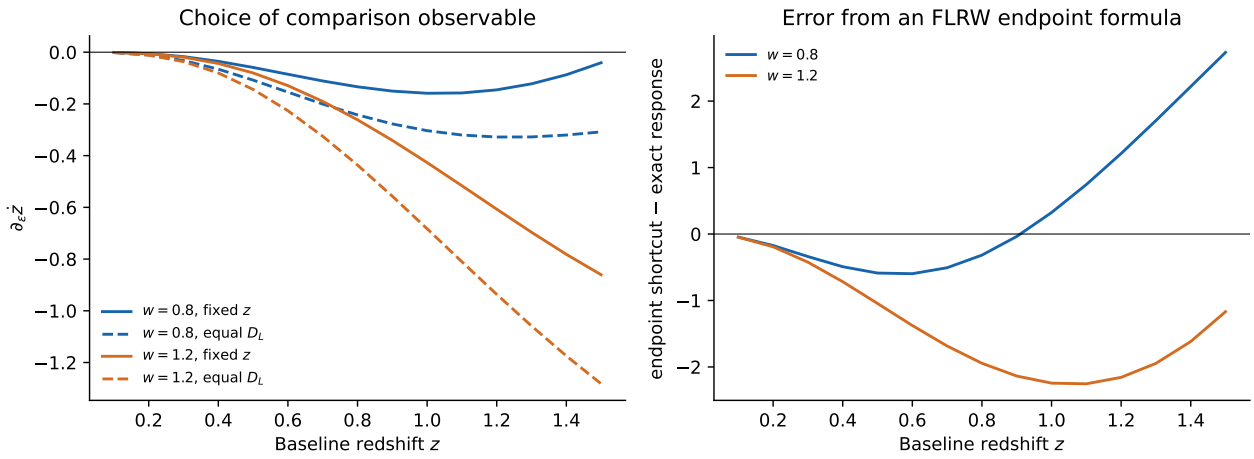


Figure 2: *Left:* Fixed-z (solid) and equal-D_L (dashed) derivatives for each bang-time width. *Right:* Derivative from the FLRW endpoint shortcut minus the correct fixed-z drift derivative. This difference can change sign with perturbation shape. Both panels use the same dimensionless parameter and baseline redshift grid.

5 Global shell-crossing test and a regular bang-time family

For the parabolic solution of Eq. (11), $R' = \tau^{-1/3}[\tau - 2rt'_{\rm B}(r)/3]$. For an expanding shell with $r > 0$, $t'_{\rm B} > 0$ therefore forces $R' = 0$ at $\tau = 2rt'_{\rm B}/3 > 0$, a shell crossing after the local bang. The diagnostic hump $f_w = r^2 e^{-(r/w)^2}$ has $f'_w > 0$ for $0 < r < w$ and $f'_w < 0$ for $r > w$. Thus *either nonzero sign* of ϵ produces shell crossings somewhere in the unbounded model. The negative curves in Figs. 1 and 2 are valid computations on their sampled light cones but are not globally regular cosmologies.

Consider instead the centrally smooth, monotone family

$$t_{\rm B}(r) = \eta f_w^-(r), \qquad f_w^-(r) = -[1 - e^{-(r/w)^2}], \qquad \eta \geq 0, \quad w \in \{0.8, 1.2\}. \tag{14}$$

Here $t'_{\rm B} \leq 0$, hence $R' = \tau^{2/3} - 2rt'_{\rm B}/(3\tau^{1/3}) > 0$ at every $r > 0$ and $\tau > 0$. This proves absence of shell crossings in the expanding parabolic solution, though it does not establish observational viability. The exact variational equation yields a *positive* equal-distance derivative at all 15 sampled redshifts: $[0.002324, 2.152592]$ for $w = 0.8$ and $[0.001042, 1.546575]$ for $w = 1.2$. Richardson extrapolation from $\eta = 0.001$ and 0.0005 agrees with the variational result within 2.93×10^{-7} and 1.36×10^{-7}, respectively. The original negative response is therefore not stable under an admissible change of perturbation family; neither family has a sampled zero crossing.

6 An exact matter-plus-Λ background

The sign check is repeated in a flat Λ-LTB subclass with $H_0 = 1$, $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $M(r) = \Omega_m r^3/2$, $E = 0$, and $\Lambda = 3\Omega_\Lambda$. Its exact expanding solution is $R(t, r) = ra(t - t_{\rm B}(r))$, where

$$a(\tau) = \left(\frac{\Omega_m}{\Omega_\Lambda}\right)^{1/3} \sinh^{2/3}\left(\frac{3}{2}\sqrt{\Omega_\Lambda}\,\tau\right), \quad t_0 = \frac{2\,\mathrm{arsinh}\,\sqrt{\Omega_\Lambda/\Omega_m}}{3\sqrt{\Omega_\Lambda}} = 0.964099\ldots. \tag{15}$$

For Eq. (14), $R' = a - rt'_{\rm B}\dot{a} > 0$ for every expanding post-bang shell. We use $\dot{R}' = \dot{a} - rt'_{\rm B}\ddot{a}$ and $\ddot{R}' = \ddot{a} - rt'_{\rm B}\dddot{a}$ in Eqs. (3) and (4). The homogeneous limit reproduces

$$\frac{Z_{\Lambda\rm CDM}(z)}{H_0} = (1 + z) - \sqrt{0.3(1 + z)^3 + 0.7}. \tag{16}$$

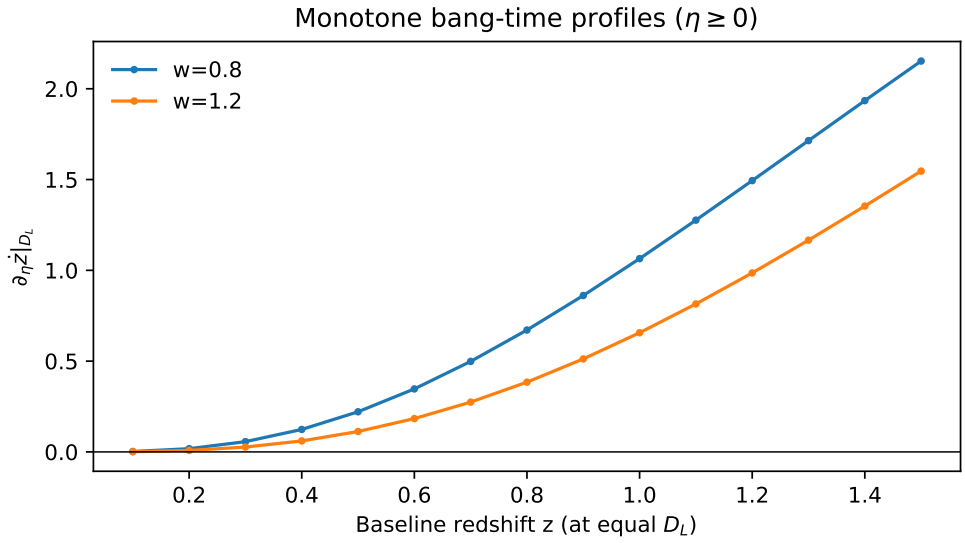


Figure 3: Equal-distance drift derivative for the shell-crossing-free parabolic families in Eq. (14). The derivative is taken at $\eta = 0$ from the physical $\eta \geq 0$ side. Positive values at the plotted points do not constitute a theorem for all profiles or redshifts.

The maximum absolute error against Eq. (16) is 2.85×10^{-11} on the grid; neighboring-ray differencing agrees with propagated drift within 3.40×10^{-11}. The equal-distance responses remain positive at all 15 points, ranging from 0.000826 to 2.127014 ($w = 0.8$) and 0.000369 to 1.440526 ($w = 1.2$), in units of $\partial_\eta(Z/H_0)$. The two-step positive-amplitude extrapolations differ by at most 2.84×10^{-6} and 6.55×10^{-7}. Figure 4 shows the calculation. This calibrated *background* is not a fitted inhomogeneous void.

7 Functional distance compensation in the dust model

To answer the stronger fixed-distance-*curve* question, set $2E(r) = \eta r^2 e(r)$ while holding $M = 2r^3/9$ and $t_{\rm B} = \eta f(r)$. To first order about EdS, the exact evolution equation gives

$$R(t,r) = r\left[t^{2/3} + \eta\left(\frac{9}{20}e(r)t^{4/3} - \frac{2}{3}f(r)t^{-1/3}\right)\right] + O(\eta^2). \tag{17}$$

This makes the curvature (growing) and bang-time (decaying) variations explicit. Write $u_t = \partial_\eta t|_r$, $u_z = \partial_\eta z|_r$, and $q = 1 - r/3$ on the EdS null cone. Direct substitution into $\delta D_L|_z = 0$ reduces the entire distance constraint to

$$Q(r) \equiv \frac{2r}{3q^5}[u_t - f(r)] + \frac{3}{2}(r-1)u_z + \frac{9r}{20}e(r) = 0. \tag{18}$$

At every $0 < r < 1$ this equation algebraically determines e in terms of the ray variations. Substituting it and its derivative into the variational light-cone equations yields a local initial-value problem. Its coefficient multiplying the solved-for e' is $(r-1)/q$; hence the inversion is singular at $r = 1$, where the EdS angular-diameter distance has its maximum ($z = 1.25$). A smooth continuation requires an additional compatibility condition there, consistent with known LTB inverse-problem regularity issues [4].

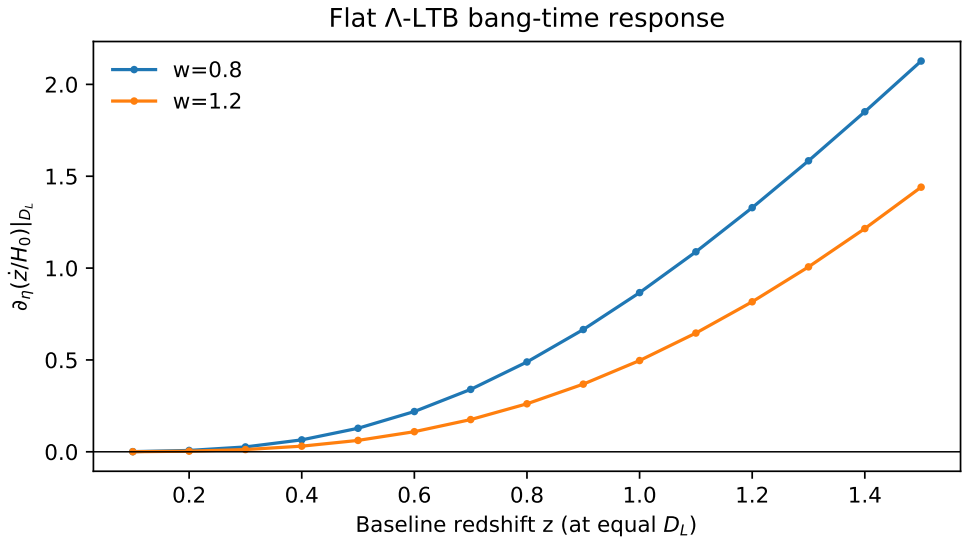


Figure 4: Equal-distance response of the exact flat Λ-LTB bang-time family. η is a dimensionless bang-time amplitude in units H_0^{-1}; the plotted derivative is of Z/H_0. No observational errors or amplitude constraints are applied.

For the monotone profiles, a center-regular numerical integration to $r = 0.98$ enforces $|Q| < 6\times10^{-14}$ on the evaluated grid, but the required $e(r)$ grows from 32.58 at $r = 0.9$ to 657.80 at $r = 0.98$ for $w = 0.8$ (and 18.59 to 376.28 for $w = 1.2$). Moving the numerical start radius from 10^{-5} to 10^{-6} changes the values at $r = 0.9$ by less than 2×10^{-3}. Figure 5 shows this rapid growth and the associated drift response. Thus a *local* first-order distance compensation exists, but these center-regular solutions do not yield a controlled smooth compensation through the distance maximum in this fixed-M gauge. Near that point $|\eta e| \ll 1$ fails for any fixed nonzero η, so the first-order curve cannot be promoted to a finite-amplitude prediction there. We do not claim a general impossibility theorem for other bang-time functions or gauges.

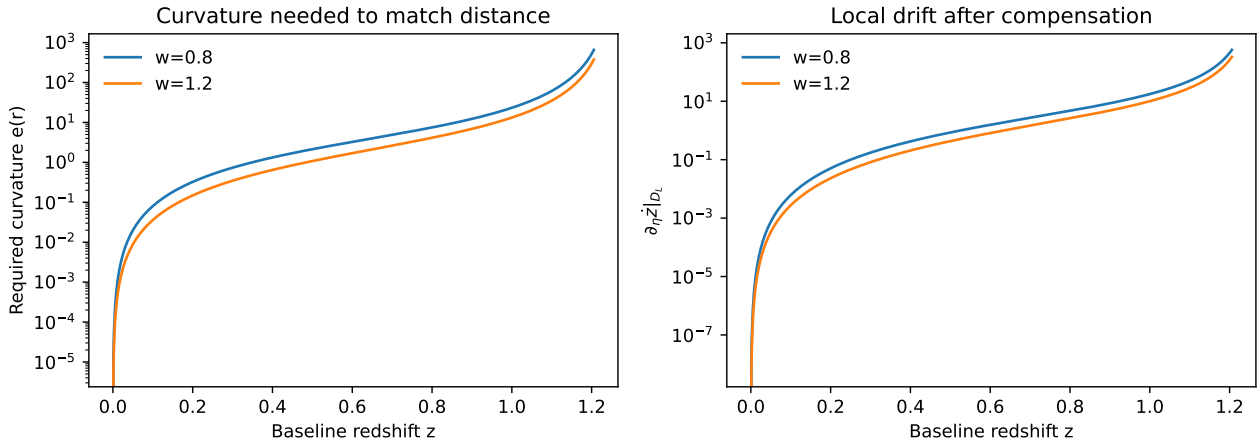


Figure 5: Local linear curvature reconstruction enforcing $\delta D_L(z) = 0$ for the monotone profiles. Logarithmic vertical axes expose rapid growth toward the EdS distance maximum at $z = 1.25$; integration stops at $r = 0.98$ ($z = 1.206$). Large derivatives near the endpoint violate finite-amplitude perturbative control.

8 Answers and observational limits

Within the stated families, the source drift is Eq. (4), and the fixed-z and equal-D_L responses differ as Eq. (9) and Fig. 2 demonstrate. The diagnostic hump profiles have shell crossings and a negative sampled response; monotone shell-crossing-free profiles give a positive sampled response in both EdS and flat Λ backgrounds. No profile-only universal zero crossing follows. Requiring the *whole* distance curve to agree is more restrictive than matching one distance: the explicitly reconstructed curvature mode develops a singular inverse problem near the angular-distance maximum for our profiles.

Global absence of shell crossing does not imply consistency with the CMB, kSZ effect, supernovae, or structure formation. Earlier analyses of general varying-bang-time voids found strong combined constraints [5]; significant decaying modes in void models that mimic acceleration are also constrained by CMB spectral distortions [6]. Those results do not supply a numerical bound for the Λ-LTB profiles specified here without fitting them. Our background calibration is not a data likelihood, and the dimensionless derivatives are not a forecast for a telescope. The concrete conclusion is a reproducible, profile-dependent response and a demonstrated inversion obstruction in the tested gauge, not a detected signal.

Reproducibility

The accompanying `analysis.py`, `viable.py`, `lambda_model.py`, and `constrained.py` generate the numerical results and vector figures. Run each script, then `pdflatex paper.tex` twice. Machine-readable outputs record all grid values. The independent checks cover EdS and flat ΛCDM limits, neighboring observer times, positive-amplitude convergence, and the variational response. The constrained inverse checks its distance identity and center-start stability. No observed data or generated artwork enters the figures.

A Explicit forcing of the bang-time variation

For the profile of Eq. (10), define $g(r) = f_w(r) + r f_w'(r)$. Holding t, r fixed and differentiating Eq. (11) at $\epsilon = 0$ gives

$$A_\epsilon = -\frac{2g}{3t^{1/3}}, \qquad B_\epsilon = \frac{2g}{9t^{4/3}}, \qquad C_\epsilon = -\frac{8g}{27t^{7/3}}. \tag{19}$$

The baseline time derivatives needed in Eq. (7) are

$$A_t = \frac{2}{3t^{1/3}}, \qquad B_t = -\frac{2}{9t^{4/3}}, \qquad C_t = \frac{8}{27t^{7/3}}. \tag{20}$$

We integrate Eq. (7) with these forcing terms along Eq. (13), then apply Eqs. (8) and (9). This analytically differentiated initial-value problem provides a check independent of changing ϵ in the nonlinear ray integrator. Agreement on the sampled interval tests the implementation; it does not guarantee convergence at arbitrarily high redshift or global absence of shell crossings.

B Coefficient of the distance-inversion equation

Solving Eq. (18) for the curvature perturbation gives

$$e = -\frac{40}{27q^5}(u_t - f) - \frac{10(r-1)}{3r}u_z. \tag{21}$$

At fixed r, its derivatives with respect to the ray variations are $e_{u_t} = -40/(27q^5)$ and $e_{u_z} = -10(r-1)/(3r)$. In the first two variational equations, the coefficients of e' are respectively $G_t = -9rq^4/20$ and $G_z = 3r/(5q)$. Implicit differentiation of Eq. (21) therefore puts the coefficient

$$1 - e_{u_t}G_t - e_{u_z}G_z = \frac{r-1}{q} \tag{22}$$

in front of e'. This vanishes at the EdS angular-distance maximum. The reconstruction script evaluates the full right-hand side before division and records the resulting local solution; it does not numerically step across $r = 1$.

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